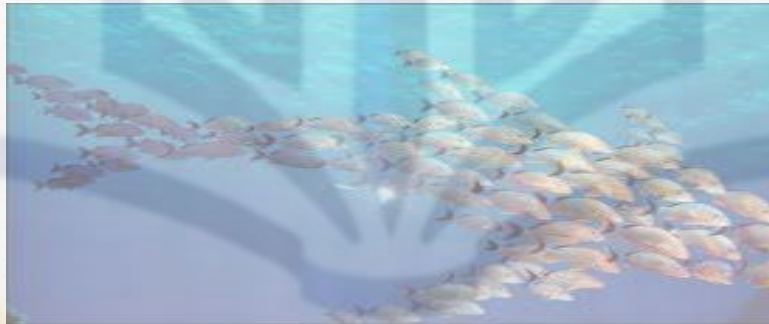


Major group of living Fishes

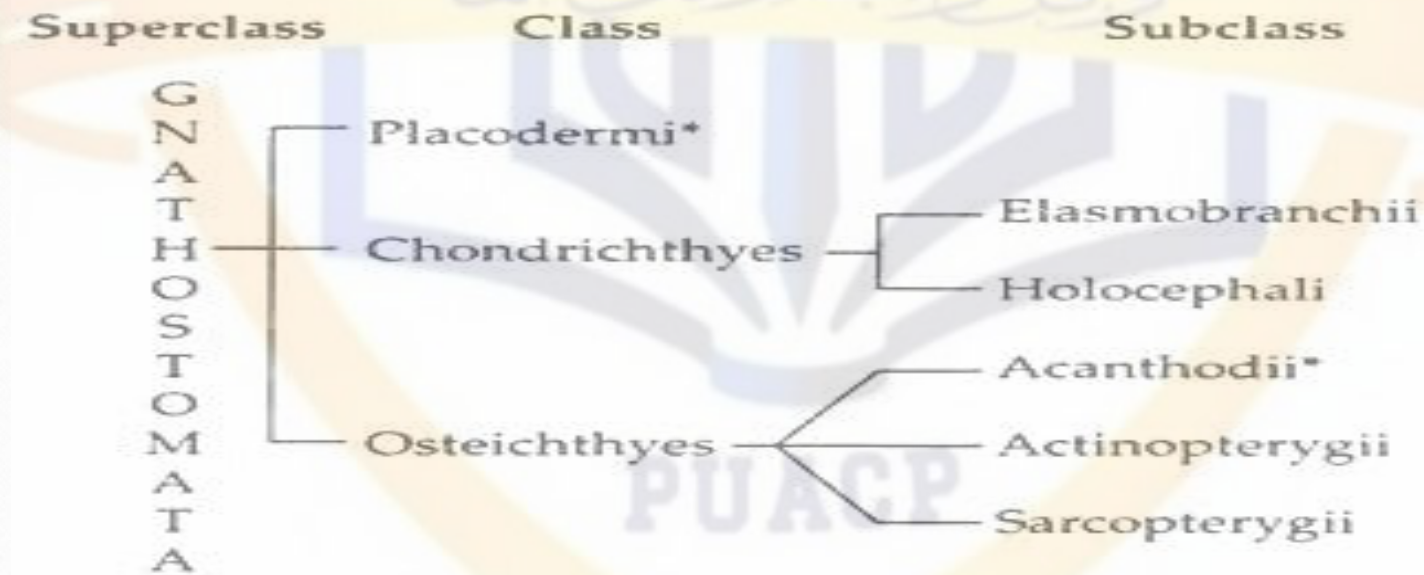


Introduction

Living fishes with jaws mostly fall into two well marked classes, the cartilaginous fishes (**Chondrichthyes**) and the bony fishes (**Osteichthyes**) including the familiar ray- finned fishes and the lung fishes. These two groups arose in the late Devonian period of geological time scale. Before that time various other types of fish dominated the waters.

Classification

- Classificatory scheme of fishes by J. Z. Young (1981) is shown:



Class Placodermii

Their epidermis were covered by heavy bony plates. Instead of gill slits they possessed spiracles for respiration. e.g. dinichthyes

PLACODERMI FISHES



DINICHTHYES

PUACP

Class Chondrichthyes

- **General characters:**
- Most of the members are marine and are carnivorous.
- Epidermis is covered by closely set placoid scales.
- Endoskeleton is cartilaginous.
- Median and paired fins are present. Fins are supported by cartilaginous fin rays.
- A spiral valve is present in the intestine.
- Swim bladder is absent.

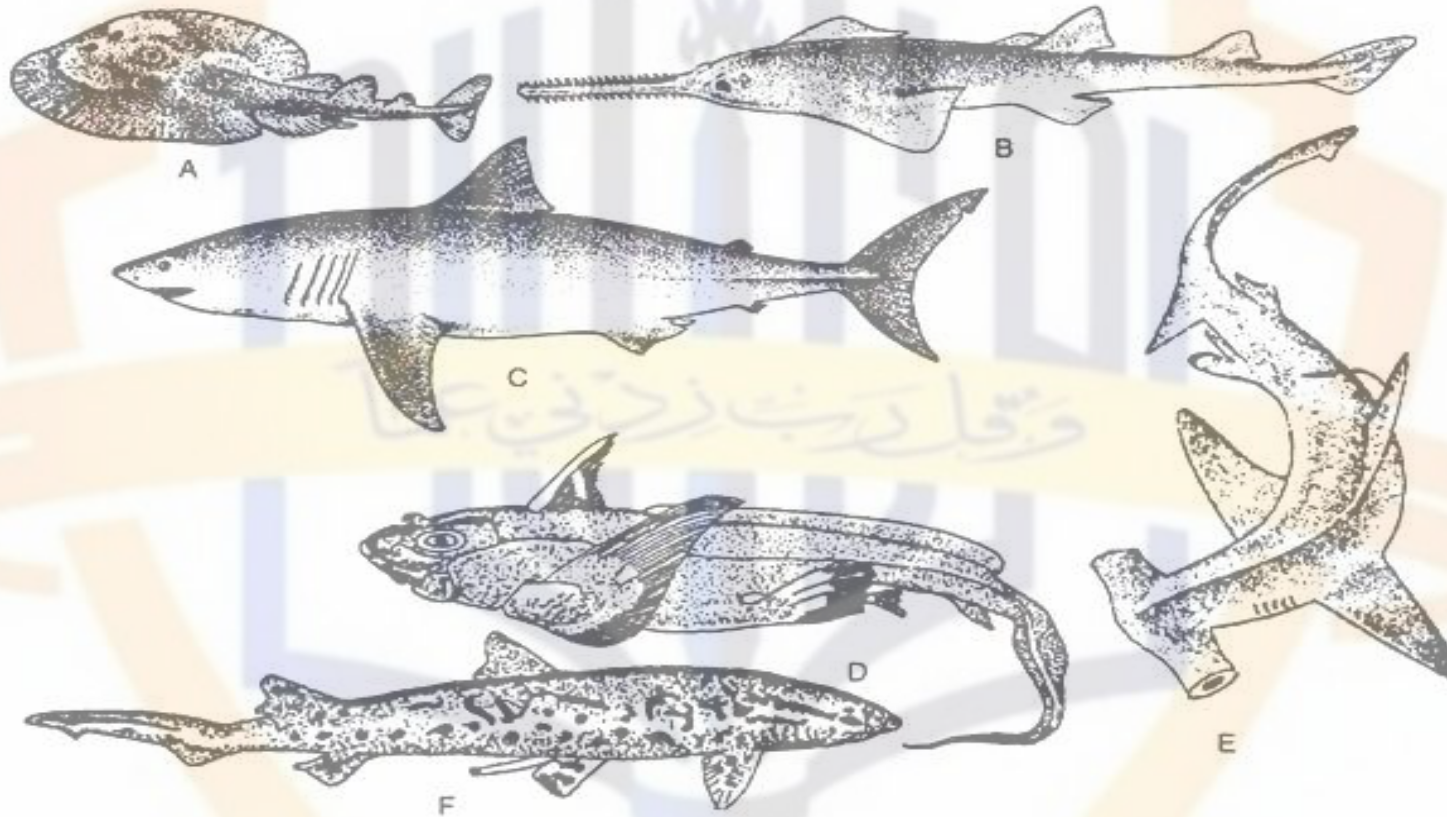
Cont..

- Generally operculum is absent in these fishes, instead, gill slits are present.
- Mouth is sub-terminal in position.
- Jaws are armoured with rows of sharp teeth. These teeth are modified placoid scales.
- Caudal fin is of heterocercal type, i.e. two lobes of the fin are unequal.
- Unisexual fish, males having clasper for reproduction. Clasper is the modified organ from pelvic fins.
- Internal fertilization is the characteristic of these fishes.

Subclass Elasmobranchii

General characters:

- At the anterior end of the body 5 to 7 gill slits are present in each side.
- Mouth is sub-terminal in position.
- These fishes can swim with high speed.
- Firm and hard jaws are provided with numerous sharp teeth.
- Brain is developed. Powerful olfactory system helps in detection of food.
- There are some ribs in the thoracic cavity.



A. (Torpedo), **B.** (Saw fish), **C.** (Carchardon), **D.** (Chimaera) & **E.** (Hammer headed Shark)

Subclass Holocephali

General characters:

- 1. Small mouth aperture is guarded by jaws and lips.
- 2. Plate like teeth firmly attached with the jaws. Holostylic upper jaw, i.e. rigidly attached to the skull.
- 3. Gill slit partially covered by operculum.
- 4. Cloaca absent, i.e., anus and urinogenital aperture separate.
- 5. Skin is naked in adults.
- 6. In addition to pelvic clasper males possess another clasper on the head.
- 7. Spiracle absent.

HOLOCEPHALI FISH

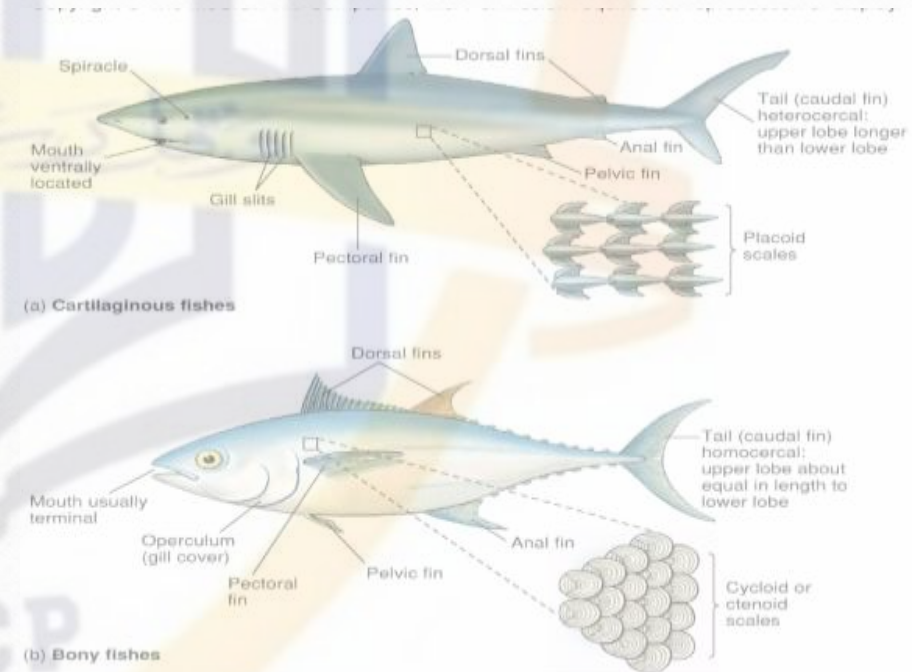


CHIMAERA

Example
D. Chimaera (Rat fish)

Class Osteichthyes

- Bony fish
- Largest group of living vertebrates
- Gills covered by operculum
- Usually swim bladder
- Highly maneuverable fins
- Cycloid scales



Cont..

1. Subclass Sarcopterygii
2. Subclass Actinopterygii



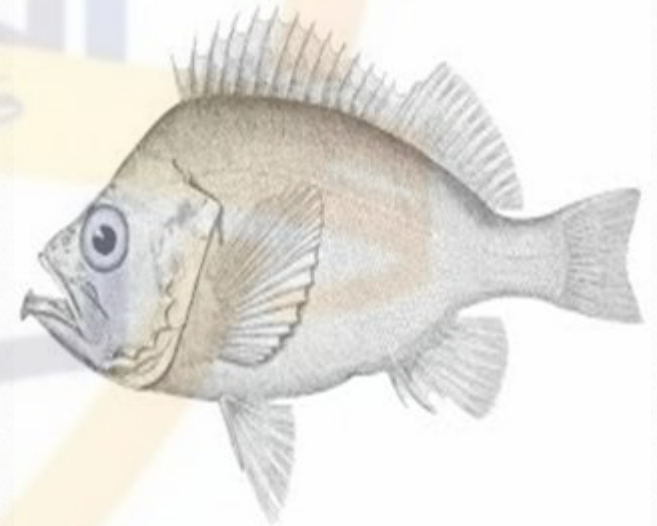
Subclass Sarcopterygii

- lobed – fin finned
- Fins supported by main axis of bone
- Modified swim bladder
- Breathe air
- Lungfish

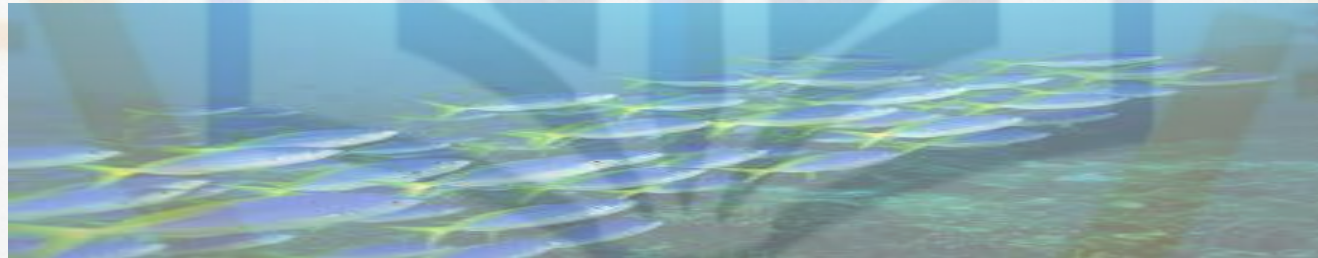


Class Actinopterygii

- Ray – finned fishes
- Generally spiracle is absent
- Marine or freshwater habitat
- Endoskeleton is either bony or cartilaginous



Characterization of living fish groups



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History of fish

Fishes are first evolved in Silurian period of Paleozoic era. They increased their number in Devonian period. Devonian Period is Called Age of Fishes.

Fishes are first jawed vertebrates. They live in water. They are cold blooded vertebrates.



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General characters of fish

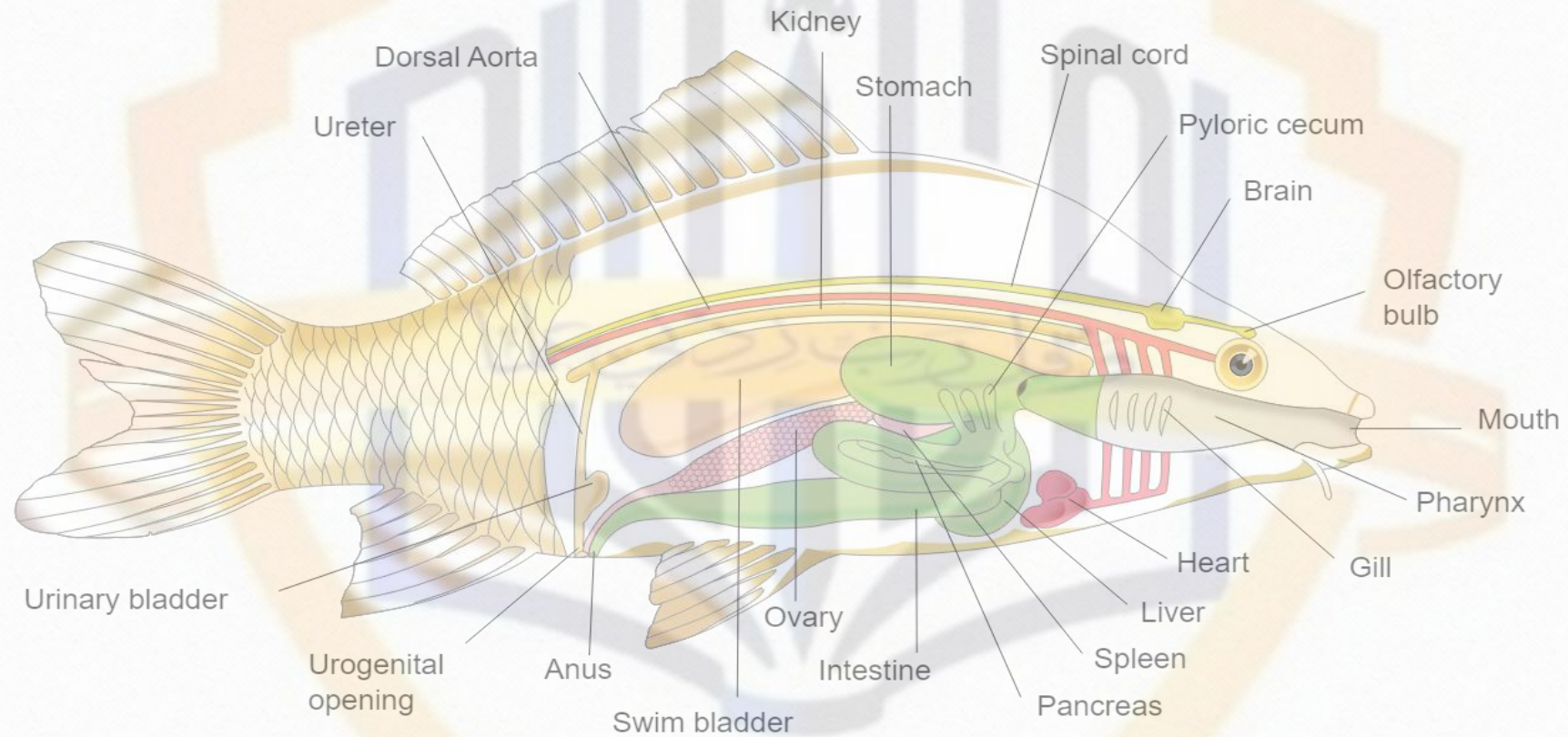
- Fishes are aquatic; **cold blooded** vertebrates
- Their body is divisible into head, trunk and tail. Neck is absent
- It has a **spindle shaped body**. It is helpful in swimming
- The body is covered by **scales**. They are placoid scales, cycloid scales, ctenoid scales, ganoid scales etc.
- Respiration is by **gills**. Gills are the extensions of the pharynx. In the elasmobranches fishes, the gills will open separate. In bony fishes the gill slits are covered by **operculum**
- On the head a **pair of nostrils** are present internal nostril are absent. In Dipnoi (lungfish) Internal nostrils are present

Cont..

- On the head a **pair of eyes** is present
- On the lateral sides of the body **LATERAL LINE SENSE** organs are present. They detect the pressure changes of water
- The body shows **paired and unpaired fins**. Pelvic and pectoral fins are paired. Dorsal and ventral fins are unpaired. They maintain balance in water. They are useful for locomotion
- The **digestive system** is well developed. In the intestine of shark scroll valve is present. The nervous system contains brain and spinal cord. Brain is small & it will not occupy the entire cranial cavity
- **10 pairs of cranial nerves** are present
- **Kidneys** are mesonephric

Cont..

- **Urinary bladder** is absent
- The **skeleton** of some fishes is made by cartilage. They are called cartilage fishes. In some fishes the skeleton is made by bone. They are called bony fishes
- In some fresh water fishes accessory **respiratory organs** are present. They will take up aerial respiration
- **Sexes** are separate. In male sharks claspers are present
- Many fishes are **oviparous**. Some fishes are **viviparous**
- In the development of fishes amnion is not developed. Hence these animals are called **anamniotes**



External & Internal Characteristics of Fish

Major groups of extinct fishes



Extinct Species

“A species which is no longer present in its original range or as a distinct species any longer is called extinct species. It was previously living and existent but has now vanished from Earth.”

As of **September 2016**, the International Union for Conservation of Nature (IUCN) lists **65** extinct fish species, **87** possibly extinct fish species, and six extinct in the wild fish species.

Groups of extinct Fishes

- Salmoniformes
- Cyprinodontiformes
- Cypriniformes
- Gasterosteiformes
- Osmeriformes
- Perciformes

Salmoniformes

Extinct Species

- Longjaw cisco (*Coregonus alpenae*)
- *Coregonus bezola*
- *Coregonus gutturosus*
- Gravenche (*Coregonus hiemalis*)
- Deepwater cisco (*Coregonus johannae*)
- Blackfin cisco (*Coregonus nigripinnis*)

Cyprinodontiformes

Extinct species

- Parras characodon (*Characodon garmani*)
- Santa Cruz pupfish (*Cyprinodon arcuatus*)
- Villa Lopez pupfish (*Cyprinodon ceciliae*)
- Cachorrito de la Trinidad (*Cyprinodon inmemoriam*)
- Parras pupfish (*Cyprinodon latifasciatus*)
- Perritos de sandia (*Cyprinodon spp.*)

Cypriniformes

Extinct species

- Beyşehir bleak (*Alburnus akili*)
- İznik shemaya (*Alburnus nicaeensis*)
- Snake River sucker (*Chasmistes muriei*)
- Mexican dace (*Evarra bustamantei*)
- Plateau chub (*Evarra eigenmanni*)
- Endorheic chub (*Evarra tlahuacensis*)

Gasterosteiformes & Osmeriformes

Extinct species

- Techirghiol stickleback (*Gasterosteus crenobiontus*)

Extinct species

- New Zealand grayling (*Prototroctes oxyrhynchus*)

Perciformes

Extinct species

- *Ctenochromis pectoralis*
- Maryland darter (*Etheostoma sellare*)
- *Ptychochromis onilahy*
- *Ptychochromoides itasy*

Class	Typical Examples		Key Characteristics	Approximate Number of Living Species
Acanthodii	Spiny fishes		Fishes with jaws; all now extinct; paired fins supported by sharp spines	Extinct
Placodermi	Armored fishes		Jawed fishes with heavily armored heads; often quite large	Extinct
Osteichthyes Actinopterygii (subclass)	Ray-finned fishes		Most diverse group of vertebrates; swim bladders and bony skeletons; paired fins supported by bony rays	30,000
Sarcopterygii (subclass)	Lobe-finned fishes		Largely extinct group of bony fishes; ancestral to amphibians; paired lobed fins	8
Chondrichthyes	Sharks, skates, rays		Streamlined hunters; cartilaginous skeletons; no swim bladders; internal fertilization	750
Myxini	Hagfishes		Jawless fishes with no paired appendages; scavengers; mostly blind, but a well-developed sense of smell	30
Cephalaspidomorphi	Lampreys		Largely extinct group of jawless fishes with no paired appendages; parasitic and nonparasitic types; all breed in freshwater	35